

Toan Nguyen

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Education

University of Southern California, PhD in Computer Science Aug 2025 – Present

- Research topics: Robotic Manipulation, In-Context Imitation Learning, Embodied Chain-of-Thought Reasoning
- Advisors: **Prof. Yue Wang** and **Prof. Daniel Seita**

Ho Chi Minh City University of Science, BS in Computer Science Sep 2019 – March 2024

- GPA: 3.64/4.00
- Thesis: Open-Vocabulary Affordance Detection in 3D Point Clouds
- Advisor: **Prof. Bac Le**

Research Experience

AI Research Resident, FPT Software AI Center Jul 2022 – Apr 2025

- Robotics and Computer Vision Team. Advisors: **Prof. Anh Nguyen** and **Prof. Ngan Le**
- Research topics: Multimodal Robotic Manipulation, Generative Models for Robotics and 3D Scene Generation, Embodied Navigation, Robot Design Understanding, Multi-Agent Systems, Identity-Preserving Image Generation

Undergraduate Research Assistant, AISIA Research Lab Jan 2022 – Jul 2022

- AI for Life Sciences Team. Advisor: **Prof. Binh Nguyen**
- Research topics: Representation Learning, Image Data Augmentation, Insect Pest Classification and Detection

Publications

Rethinking Progression of Memory State in Robotic Manipulation: An Object-Centric Perspective

Nhat Chung*, Taisei Hanyu*, **Toan Nguyen**, Huy Le, Frederick Bumgarner, Duy Nguyen Ho Minh, Khoa Vo, Kashu Yamazaki, Chase Rainwater, Tung Kieu, Anh Nguyen, Ngan Le
AAAI Conference on Artificial Intelligence (AAAI), 2026

Language-Driven 6-DoF Grasp Detection Using Negative Prompt Guidance

Toan Nguyen, Minh N. Vu, Baoru Huang, An Vuong, Quan Vuong, Dung Nguyen, Ngan Le, Thieu Vo, Anh Nguyen
European Conference on Computer Vision (ECCV), 2024. **Oral Presentation (2.33%)**

HabiCrowd: A High-Performance Simulator for Crowd-Aware Visual Navigation

An Vuong, **Toan Nguyen**, Minh N. Vu, Baoru Huang, Dung Nguyen, Binh Huynh, Thieu Vo, Anh Nguyen
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024

Language-Conditioned Affordance-Pose Detection in 3D Point Clouds

Toan Nguyen, Minh N. Vu, Baoru Huang, Tuan Vo, Vy Truong, Ngan Le, Thieu Vo, Bac Le, Anh Nguyen
IEEE International Conference on Robotics and Automation (ICRA), 2024

Open-Vocabulary Affordance Detection Using Knowledge Distillation and Text-Point Correlation

Tuan Vo, Minh N. Vu, Baoru Huang, **Toan Nguyen**, Ngan Le, Thieu Vo, Anh Nguyen
IEEE International Conference on Robotics and Automation (ICRA), 2024

Language-Driven Scene Synthesis Using Multi-Conditional Diffusion Model

An Vuong, Minh N. Vu, **Toan Nguyen**, Baoru Huang, Dung Nguyen, Thieu Vo, Anh Nguyen
Conference on Neural Information Processing Systems (NeurIPS), 2023

Open-Vocabulary Affordance Detection in 3D Point Clouds

Toan Nguyen, Minh N. Vu, An Vuong, Dung Nguyen, Thieu Vo, Ngan Le, Anh Nguyen

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023. **Best Overall & Best Student Paper Awards Finalist (0.43%)**

RoboDesign1M: A Large-scale Dataset for Robot Design Understanding

Tri Le, **Toan Nguyen**, Quang Tran, Quang Nguyen, Baoru Huang, Hoan Nguyen, Minh N. Vu, Tung Ta, Anh Nguyen

Under submission to ICRA 2026

SlotVLA: Towards Modeling of Object-Relation Representations in Robotic Manipulation

Taisei Hanyu*, Nhat Chung*, Huy Le, **Toan Nguyen**, Yuki Ikebe, Anthony Gunderman, Duy Nguyen Ho Minh, Khoa Vo, Tung Kieu, Kashu Yamazaki, Chase Rainwater, Anh Nguyen, Ngan Le

Under submission to ICRA 2026

FurniMAS: An LLM-Based Multi-Agent System for Language-Guided Furniture Decoration

Toan Nguyen, Tri Le, Quang Nguyen, Anh Nguyen

Under submission to EACL 2026

Presentations

Oral paper presentation, ECCV 2024

Milano, Italy

Oral paper presentation, IROS 2023

Michigan, USA

Honors and Awards

- Viterbi Graduate Fellowship, University of Southern California, 2025
- Registration Grant, ECCV, 2024
- Oral Presentation, ECCV, 2024
- Award for Technology and Innovation Publications, Ho Chi Minh City University of Science, 2024
- Best Paper & Best Student Paper Awards Finalist, IROS, 2023

Professional Services

- Conference Reviewer: ICRA 2026, WACV 2026, IROS 2025, ECCV 2024, ICRA 2024, IROS 2024
- Journal Reviewer: RA-L 2025

Skills

Programming: Python, C/C++, Julia, R, Java, JavaScript, SQL, HTML, CSS, Linux, MATLAB

Tools: Git, Google Cloud, Jupyter Notebook/Lab, Blender, Docker

Languages: English (8.0 IELTS), Vietnamese